

Track-Based Cyclone Impact & Infrastructure Vulnerability Forecaster

Architecture & Technical Reference — Problem Statement 05, Theme: Resilience

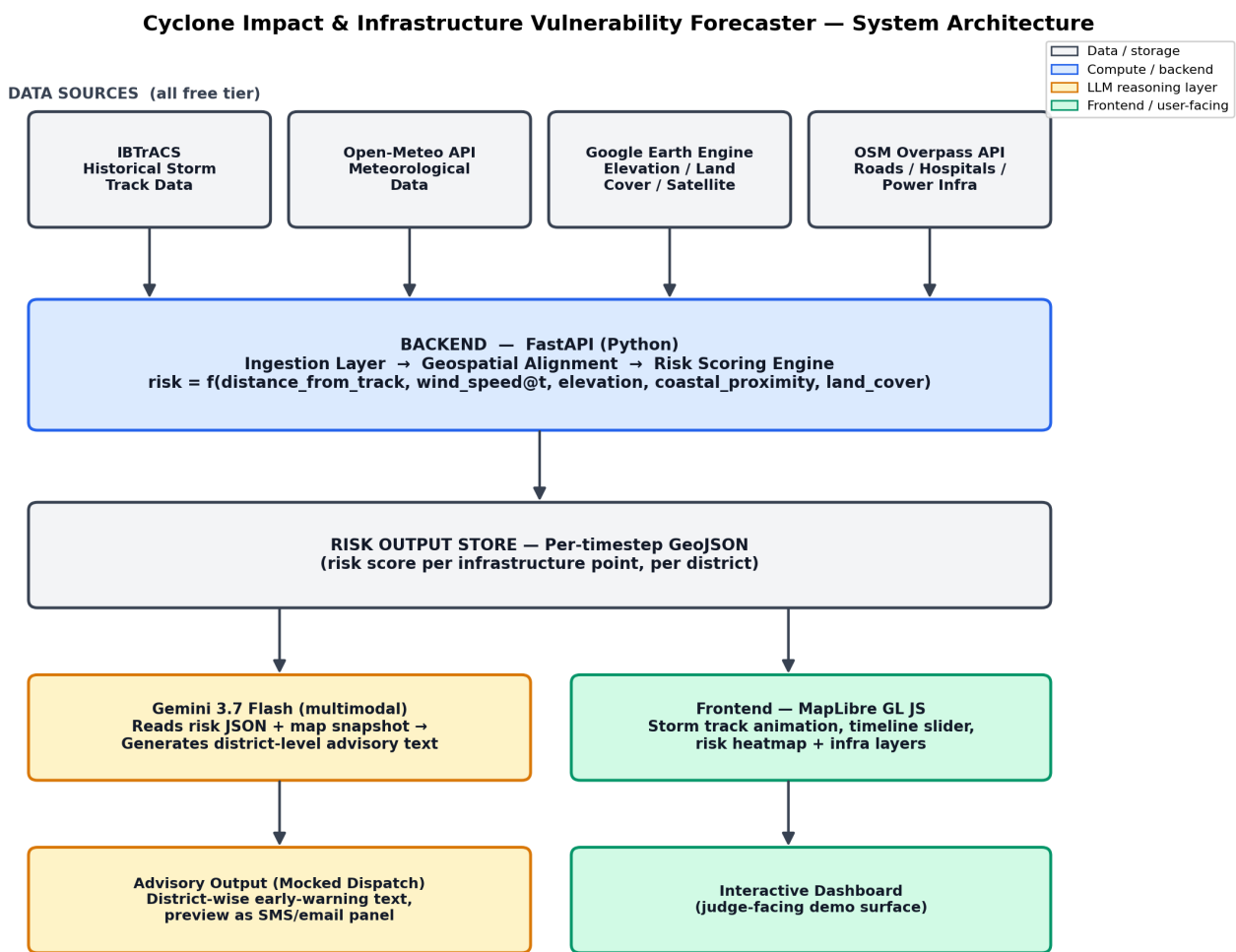
1. Problem Recap

Extreme weather events in the Bay of Bengal and coastal APAC require rapid anticipatory action. The goal is to shift disaster response from post-landfall recovery to pre-landfall evacuation planning, infrastructure hardening, and parametric insurance liquidity.

The challenge, as stated: Build an AI-powered predictive risk and vulnerability modeling platform utilizing Google Earth Engine (GEE) satellite feeds, real-time meteorological data, and Gemini 3.7 Flash's multimodal reasoning. Simulate cyclone storm surges, predict local rainfall damage pathways, map exposure for critical infrastructure (power grids, arterial roads, medical shelters), and automate early-warning advisory dispatches for local municipal and disaster management authorities.

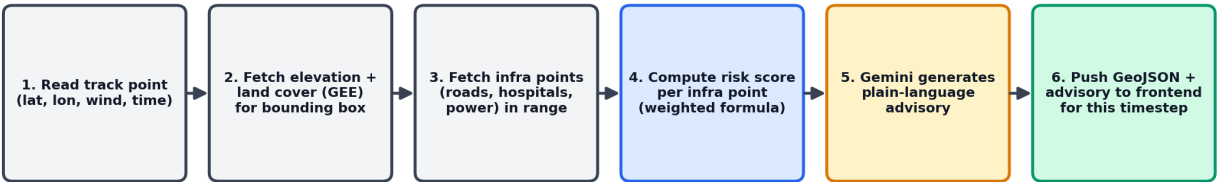
Important framing correction: This is NOT a RAG (retrieval-augmented generation) system. There is no large document corpus to retrieve from. This is a geospatial data-fusion pipeline (structured data in) feeding a generation step (advisory text out). Retrieval accuracy is not the hard problem here — data fusion and risk-score calibration is. A thin optional RAG layer can be bolted on later to ground advisory text in real NDMA/IMD protocol documents, but that is a bonus, not the core system.

2. System Architecture



3. Per-Timestep Processing Flow

Per-Timestep Processing Flow (runs once per storm-track point along the timeline)



Looped across every timestep of the storm track to animate risk evolving as landfall approaches.

4. Tech Stack & Cost Reference

Component	Tool	Purpose	Cost
Storm track data	IBTrACS	Historical cyclone tracks for demo replay (lat/lon, wind, time)	Free (no key)
Weather data	Open-Meteo API	Meteorological data for the region/period	Free, no key
Satellite / elevation	Google Earth Engine (Python API)	SRTM elevation, land cover, optical/radar imagery	Free — Community tier, no billing account
Infrastructure data	OSM Overpass API	Roads, hospitals, power infra point locations	Free, no key
LLM reasoning	Gemini 3.7 Flash (Gemini API / ML Studio)	AI-generated advisory text generation from risk factors	Free tier available, paid tier limited (~10-15 requests/min)
Backend	Python + FastAPI	Ingestion, geospatial alignment, risk scoring engine	Free
Frontend / map	MapLibre GL JS (or Leaflet)	Storm track animation, timeline slider, risk heatmap	Free
Hosting	Vercel / Render / Railway free tier	Deploy for demo	Free

5. Risk Scoring Formula (transparent, defensible in Q&A)

Keep this as an explicit weighted formula, not an opaque ML model — judges will ask how you computed it, and a formula you can explain beats an unexplainable black box.

```
risk_score(point, t) = w1 * proximity_factor(dist_from_track_at_t) + w2 *
wind_intensity_factor(wind_speed_at_t) + w3 * elevation_vulnerability(elevation_m) + w4 *
coastal_exposure(distance_to_coast) + w5 * land_cover_multiplier(land_cover_class)
```

Suggested starting weights: proximity 0.35, wind intensity 0.25, elevation 0.20, coastal exposure 0.15, land cover 0.05. Normalize each factor to 0–1 before weighting. Tune visually against the historical cyclone's known impact zones — if your replayed storm (e.g. Amphan/Remal) has documented worst-hit districts, your model should light those up first. That visual match is your credibility proof for judges.

6. Data Source Notes

- **IBTrACS** — use a single well-documented historical cyclone for the demo (Amphan 2020 or Remal 2024). Do not attempt live IMD data; there is no clean public real-time API and it adds risk for zero benefit.
- **GEE** — register for the Community quota tier (150 EECU-hours, no billing account required). Do this on day 0/hour 1, not later.
- **Overpass API** — power grid coverage in OSM is sparse for India; roads and hospitals are well covered. Be ready to supplement power infra with a small hand-curated list if Overpass returns too little.
- **Gemini 3.7 Flash** — free tier, no card required, but rate-limited per project. Call it once per district summary, not once per infrastructure point.